

Md. Abdur Rahman

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🔗 [borhanitrash](#) 🎓 [Google Scholar](#)

Education

Southeast University

BSc in Computer Science and Engineering

Oct 2021 – Nov 2025

- CGPA: 3.87/4.0
- Relevant Coursework: Data Mining, Artificial Intelligence, Image Processing

Experience

Research Intern

ELITE Lab

Queens, NY, USA

September 2025 – Present

- Working on low-resourced languages like Bangla dialects and Chakma to develop multilingual and cross-script translation systems.
- Developing LLM-based frameworks for automated medical report generation and evaluation.

Research Assistant

CAIR Lab, Southeast University

Dhaka, Bangladesh

June 2024 – October 2025

- Designed and implemented a custom CNN architecture for brain tumor classification with LIME using MRI dataset, achieving 99.89% classification accuracy, significantly outperforming standard pre-trained models.
- Designed a federated learning pipeline for multi-institutional Lung and Colon cancer classification with GradCAM, integrating a custom lightweight CNN model (380K parameters) with high accuracy of 99.94%.

Research Collaborator

Advanced Machine Intelligence Research Lab (AMIRL)

Dhaka, Bangladesh

March 2025 – Present

- Conducting an in-depth literature review for a forthcoming review paper on MRI-based brain tumor analysis, integrating recent progress in deep learning models for classification, detection, and segmentation tasks.

Publications

AlzFed-XAI: High-Fidelity Interpretable Alzheimer's Diagnosis with Privacy-Preserving Federated Learning (Accepted)

[Paper] [\[Code\]](#) [\[Code\]](#)

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum

5th Muslims in ML Workshop co-located with NeurIPS 2025 [\[Code\]](#)

ChakmaBridge: A Five-Way Parallel Corpus for Navigating the Script Divide in an Endangered Language (Accepted)

[Paper] [\[Code\]](#) [\[Code\]](#)

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum

Second Bangla Language Processing Workshop co-located with AACL-IJCNLP [\[Code\]](#)

BhasaBodh: Bridging Bangla Dialects and Romanized Forms through Machine Translation (Accepted)

[Paper] [\[Code\]](#) [\[Code\]](#)

Md. Tofael Ahmed Bhuiyan, Md. Abdur Rahman, Abdul Kadar Muhammad Masum

Second Bangla Language Processing Workshop co-located with AACL-IJCNLP [\[Code\]](#)

Detecting Manipulation in Ukrainian Telegram: A Transformer-Based Approach to Technique Classification and Span Identification

[Paper] [\[Code\]](#) [\[Code\]](#)

Md. Abdur Rahman, Md Ashiqur Rahman

Proceedings of the Fourth Ukrainian Natural Language Processing Workshop (UNLP 2025), Association for Computational Linguistics. [\[Code\]](#)

MRAViT-XAI: A Novel Multi-Resolution Attention Vision Transformer Framework with Explainable AI for Enhanced Lung and Colon Cancer Classification [\[Paper\]](#) [🔗](#) [\[Code\]](#) [🔗](#)

Md. Abdur Rahman, Sabik Aftahee, Lamim Zakir Pronay, Md Ashiqur Rahman

[IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking \(QPAIN\) 🔗](#)

SmolLab_SEU at BEA 2025 Shared Task: A Transformer-Based Framework for Multi-Track Pedagogical Evaluation of AI-Powered Tutors [\[Paper\]](#) [🔗](#) [\[Code\]](#) [🔗](#)

Md. Abdur Rahman, Md. Al Amin, Sabik Aftahee, Muhammad Junayed, Md. Ashiqur Rahman

[Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications \(BEA 2025\), Association for Computational Linguistics. 🔗](#)

SmolLab_SEU at MAHED 2025: Do Arabic-Native Encoders Surpass Multilingual Models in Detecting the Nuances of Hope, Hate, and Emotion? [\[Paper\]](#) [🔗](#) [\[Code\]](#) [🔗](#)

Md. Abdur Rahman, Md Sabbir Dewan, Md. Tofael Ahmed Bhuiyan, Md Ashiqur Rahman

[Proceedings of the Third Arabic Natural Language Processing Conference \(ArabicNLP 2025\), Empirical Methods in Natural Language Processing \(EMNLP 2025\). 🔗](#)

FedAvg-ResNet50: Privacy-Preserving Explainable Acute Leukemia Classification (Accepted & Presented) [\[Paper\]](#) [🔗](#)

Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Zakir Hossain, Md. Fokrul Islam Khan, Abdul Kadar Muhammad Masum

[5th International Conference on Applied Intelligence and Informatics \(AII 2025, USA\) 🔗](#)

FAKD-XAI: Feature-Aligned Knowledge Distillation with Explainable AI for Efficient Brain Tumor Classification (Accepted & Presented) [\[Paper\]](#) [🔗](#) [\[Code\]](#) [🔗](#)

Md. Abdur Rahman, Sabik Aftahee, Md Ashiqur Rahman, Lamim Zakir Pronay

[International Conference on Data Science, AI and Applications \(ICDSAIA\) 🔗](#)

PneumoniaXAttnNet: An Attention-Driven Xception Framework for Accurate Pneumonia Classification with XAI (Accepted) [\[Paper\]](#) [🔗](#)

Tanvir Ahmed, *Md. Abdur Rahman*, Md. Delower Hossain, Yousuf Howlader, Md. Mahi Uddin, Md. Tajuddin, MD Illeas Hossain, Md. Mijanur Rahman

[2nd IEEE Conference on Computing Applications and Systems \(COMPAS 2025\) 🔗](#)

CerviTrans-XAI: An Explainable Vision Transformer Ensemble for Accurate Cervical Cancer Classification (Accepted) [\[Paper\]](#) [🔗](#)

Md. Delower Hossain, Md. Mijanur Rahman, *Md. Abdur Rahman*, Tanvir Ahmed, Sujay Dhar, Or nab Biswass

[2nd IEEE Conference on Computing Applications and Systems \(COMPAS 2025\) 🔗](#)

Advancing Military Cybersecurity: A Scalable Ensemble Framework with SHAP Analysis for Military Operations (Accepted) [\[Paper\]](#) [🔗](#)

Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Prof. Dr. Abdul Kadar Muhammad Masum, Arif Hossen, Md Ruhul Rabbi

[6th IEEE International Conference on Telecommunications and Photonics \(ICTP 2025\) 🔗](#)

DeepSea-Net: A YOLO Based Framework for Real-Time Detection and Classification of Underwater Plastic Pollution (Accepted) [\[Paper\]](#) [🔗](#)

Progga Parmita Roy, Sabbir Sarder Munna, Md. Sazed Ul Karim, *Md. Abdur Rahman*, Md. Tofael Ahmed Bhuiyan, Md. Joseb Siddique Siam

[2025 IEEE 7th International Conference on Sustainable Technologies for Industry 5.0 \(STI 2025\) 🔗](#)

SmolLab_SEU at CheckThat! 2025: How Well Do Multilingual Transformers Transfer Across News Domains for Cross-lingual Subjectivity Detection? (Accepted & Presented)

[\[Paper\]](#) [🔗](#) [\[Code\]](#) [🔗](#)

Md. Abdur Rahman, MD AL AMIN, Md Sabbir Dewan, Md Jahid Hasan, Md Ashiqur Rahman

[CheckThat! Lab Workshop, 16th Conference and Labs of the Evaluation Forum](#) [🔗](#)

Hintervelt@LT-EDI 2025: A Transformer-Based Approach for Identifying Racial Hoaxes in Code-Mixed Hindi-English Social Media Narratives (Accepted & Presented)

[\[Paper\]](#) [🔗](#) [\[Code\]](#) [🔗](#)

Md. Abdur Rahman, MD AL AMIN, Sabik Aftahee, Md Ashiqur Rahman
[LT-EDI Workshop, 5th Conference on Language, Data and Knowledge](#) [🔗](#)

Hintervelt@LT-EDI 2025: A Transformer-Based Detection of Caste and Migration Hate Speech in Tamil Social Media (Accepted & Presented)

[\[Paper\]](#) [🔗](#) [\[Code\]](#) [🔗](#)

MD AL AMIN*, Sabik Aftahee*, *Md. Abdur Rahman**, Md Sajid Hossain Khan, Md Ashiqur Rahman

[LT-EDI Workshop, 5th Conference on Language, Data and Knowledge](#) [🔗](#)

*Authors contributed equally to this work.

Under Review Publications

MangoLeafNet-XAI: An Attention-Enhanced Deep Learning Architecture for Accurate and Interpretable Mango Leaf Disease Classification

1st Author

Submitted to *Future Generation Computer Systems (Q1 Journal)*

BEACON: A Novel Cross-Modal Architecture for High-Precision and Explainable Brain Tumor Classification Using Medical-Specific Attention

1st Author

Submitted to *International Journal of Intelligent Systems (Q1 Journal)*

HybridTransUNet: Fusing Vision Transformers with Attentive U-Net for Accurate and Interpretable Brain Tumor Segmentation

1st Author

Submitted to *International Journal of Intelligent Systems (Q1 Journal)*

FedMal-XAI: An Explainable Federated Vision Transformer Leveraging Knowledge Distillation for Privacy-Preserving Malaria Detection

2nd Author

Submitted to *Information Fusion (Q1 Journal)*

Super Learner: A Stacking Ensemble with Explainable AI for High-Fidelity Multi-Target ESG Forecasting

3rd Author

Submitted to *MIS Quarterly (Q1 Journal)*

Quantifying Corporate Resilience: How Governance Attributes and ESG-Stack Prediction Drive Financial Stability in a Multi-Sector Context

3rd Author

Submitted to *Applied Soft Computing (Q1 Journal)*

X-ArsenicDistill: An Efficient and Interpretable Framework for Arsenicosis Classification via Knowledge Distillation

1st Author

Submitted to *28th International Conference on Computer and Information Technology (ICCIT 2025)*

MalariaNet-XAI: An Explainable Deep Ensemble Framework for Accurate Malaria Detection

1st Author

Submitted to *28th International Conference on Computer and Information Technology (ICCIT 2025)*

SentimentFusion-Net: A Hybrid ML-DL Framework for Benchmarking ChatGPT User Opinions on Twitter

2nd Author

Submitted to *28th International Conference on Computer and Information Technology (ICCIT 2025)*

FedNetXAI: Privacy-Preserving Federated EfficientNet B4 for Multi-Class Alzheimer's Disease Classification with Explainable AI

2nd Author

Submitted to *12th International Conference on Next-generation Computing, Communication, Systems and Security (NSysS 2025)*

Honors & Professional Activities

Kaggle Dataset Grandmaster (Best Rank: 30th in the world)	Profile Link 
1st Place , Fragments of Feeling: A Contextual Emotion Challenge (2025)	GitHub Repo 
2nd Place , CalorieQuest: Predicting Energy Burn with Data (2025)	GitHub Repo 
3rd Place , DU AI Challenge (2025)	GitHub Repo 
Honorable Mention in ICPC Online Preliminary Contest 2024	Certificate Link 
Reviewer , 20th Workshop on Innovative Use of NLP for Building Educational Applications (BEA), Co-located with 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)	
Reviewer , 5th Muslims in ML Workshop, Co-located with the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)	
Reviewer , CheckThat!: Lab on Subjectivity, Fact-Checking, Claim Extraction & Normalization, and Retrieval, Co-located with 16th Conference and Labs of the Evaluation Forum (CLEF 2025)	

Professional Certification

GitHub Foundations Certification GitHub	Nov 2024 - Nov 2027 
Machine Learning Specialization Stanford University	2024 
Google Advanced Data Analytics Specialization Google	2024 
Building LLM Applications for Data Scientists and Software Engineers Maven	2025 

Skills

Programming Language: C, C++, Java, Python

Programming Framework: Pandas, NumPy, Matplotlib, Seaborn, PyTorch, HuggingFace, OpenCV

Technologies: Machine learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), large language model (LLM), Git, GitHub, Microsoft Office, LaTeX